

MotionPI System Visualization

Final Process Book

Visualization for Data Science (CS-6630)

Project: kinetictrace

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1. Overview & Motivation

This project presents an interactive web-based visualization system designed to help researchers inspect, validate, and interpret multi-stream behavioral, sensor, and reliability data from a mobile-health study (MotionPI).

The goal is to give collaborators a single visual platform where they can:

- verify ENMO-based activity detections,
- check whether PA-EMA triggers occurred during valid bouts,
- diagnose reliability issues such as battery drops, storage saturation, and disconnections,
- align participant events (surveys, toggles, system logs) with activity behavior over time.

The study involves wrist-worn MotionSenseHRV4 sensors and a custom smartphone app.

These two devices generate several independent data streams. Until now, collaborators have had to inspect CSVs separately or manually search through logs.

The visualization integrates these streams into one timeline, reducing analysis time from hours to minutes.

2. Related Work

Several visualization systems inspired the design of this project:

- mCerebrum / RADAR-base (mobile-health platforms)

These tools use layered timelines to show PPG, accelerometer, and EMA data. Their designs demonstrate the importance of multi-stream alignment, event markers, and clutter reduction, which strongly informed my layout.

- Fitbit & Apple Health Activity Views

These applications use smoothed activity curves, threshold indicators, and clean color palettes. I adopted similar principles:

- clear curves for moving averages,
- threshold lines,
- minimalistic layering.

- Scientific time-series visualization (PAA/EMA research papers)

Most use threshold bands, shaded windows, and overlayed markers to validate behavior. This motivated the shaded pre-window for PA-EMA rules and the day/night background to aid contextual interpretation.

These examples guided my choices on color, layering, symbol encoding, and focus-plus-context interaction.

3. Questions

This project is built around four core analytical questions:

1. Activity Detection:
When did the participant engage in enough physical activity to satisfy the PA-EMA triggering rule?
2. Trigger Validation:
Did PA-EMA triggers occur during valid activity bouts? If not, why?
3. Reliability Diagnosis:
Can gaps or anomalies be explained by battery drops, low storage, or disconnections?
4. Contextual Interpretation:
How do activity patterns change with day/night cycles, survey events, and system toggles?

Evolution of Questions

Initially the focus was only detecting PA-EMA bouts. During data inspection, new questions emerged:

- Why do some activity windows not trigger EMAs?
- Are failures caused by low battery or data collection being toggled off?
- What is the cadence of ENMO samples?
- How stable are wristband connections?

These new questions shaped the design toward a multi-stream view instead of a single-stream ENMO plot.

4. Data

The system integrates four data sources:

ENMO Activity

- Wristband accelerometer, ~1 Hz nominal cadence
- Magnitude → ENMO (Euclidean Norm Minus One)
- Primary metric for PA-EMA rule

Events / Logs

- Survey triggers
- Data collection toggles
- Connect/disconnect events
- Low-battery warnings
- These events are semantically important but sparse and irregular.

Battery & Storage

- Phone battery (%)
- Left/right wristband battery (%)
- Wristband storage usage (%)
- Essential for diagnosing reliability issues.

Timestamps

- All converted to America/Denver
- Important for aligning survey times, night/day shading, and activity rhythms.

Data Cleaning & Processing

- Cadence detection via median timestamp delta (≈ 2 seconds in sample).
- Event deduplication: merge events within ± 30 seconds.
- ENMO moving average: centered 7-minute window.
- PA-EMA rule: $\text{ENMO} \geq 0.1006$ for $\geq 70\%$ of samples in a window.
- Auto-matching left/right wristbands via MAC address.

5. Exploratory Data Analysis

Early exploration included:

Cadence Check

Using timestamp deltas, I verified the sampling rate varied between 1–2 seconds. This justified using a moving average instead of a fixed smoothing kernel.

Window Rule Testing

On the sample dataset, few windows satisfied the PA-EMA threshold. This matched the low-activity period shown in the raw ENMO plot.

Event Density

Events were clustered and noisy (multiple events at nearly identical timestamps). This motivated:

- 30-second deduplication,
- symbol encoding instead of color alone.

Battery Trends

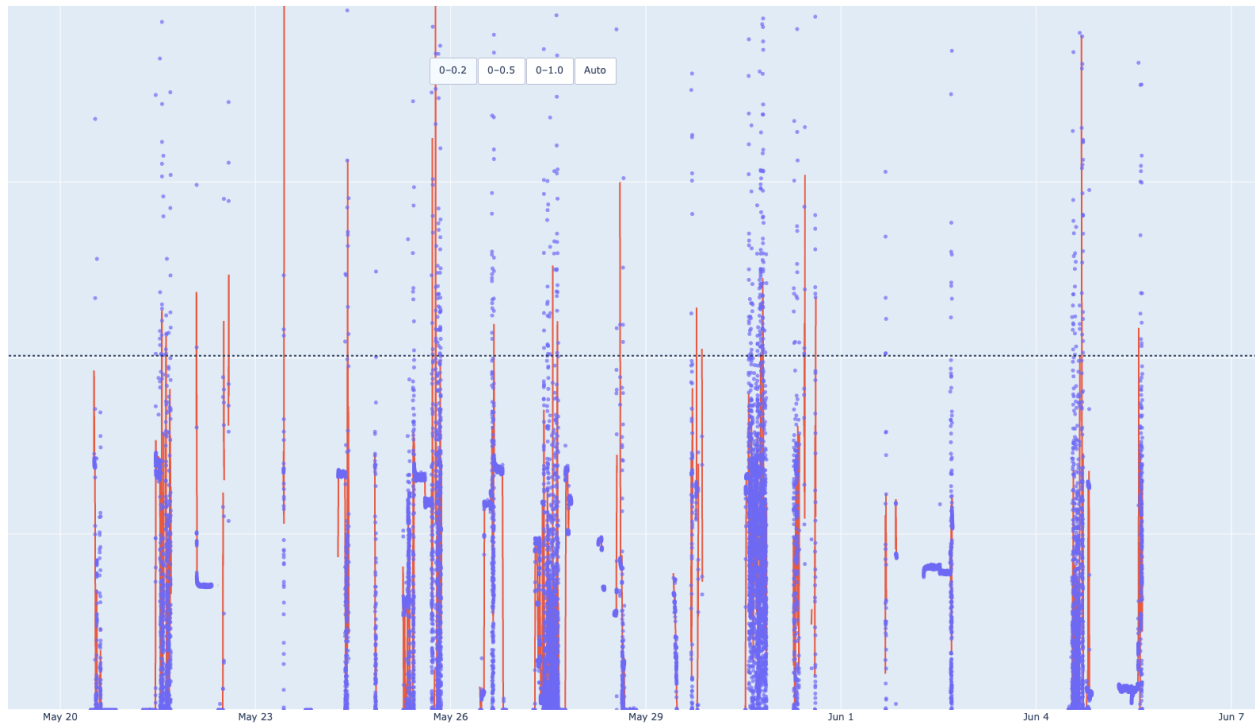
Battery and storage trends explained data gaps. Several disconnections aligned with battery dips.

These insights guided the final design toward layer toggles, downsampling, symbol redundancy, and day/night shading.

6. Design Evolution

This section describes how the visualization evolved. Three design alternatives were explored:

Design A — Multi-Row Dashboard (Rejected).



Pros

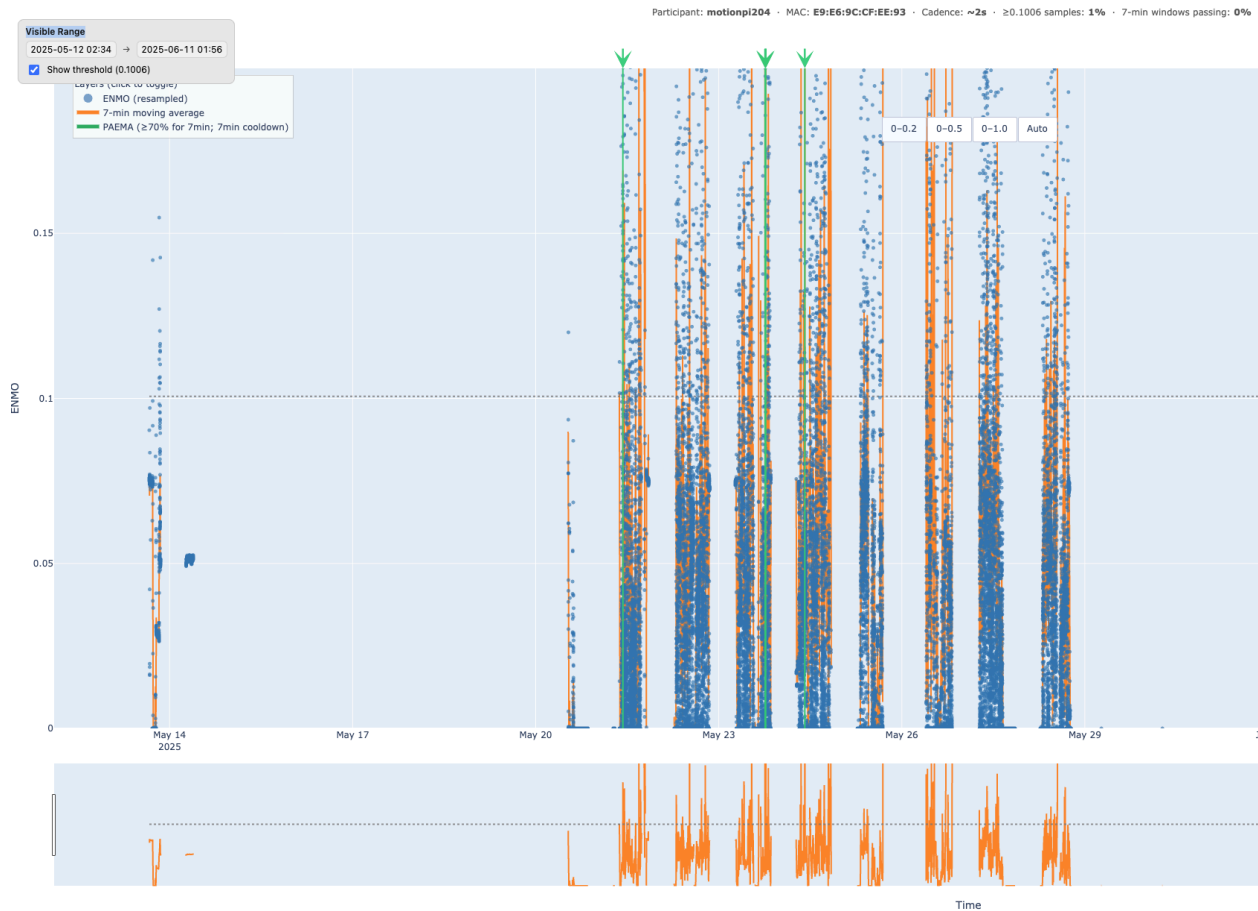
- Clean ENMO point presentation
- Easy to change vertical scale

Cons

- Hard for understand patterns and zoom issue

Rejected due to poor temporal alignment readability.

Design B — Layered Single Timeline + Toggles (Accepted)



This final design uses:

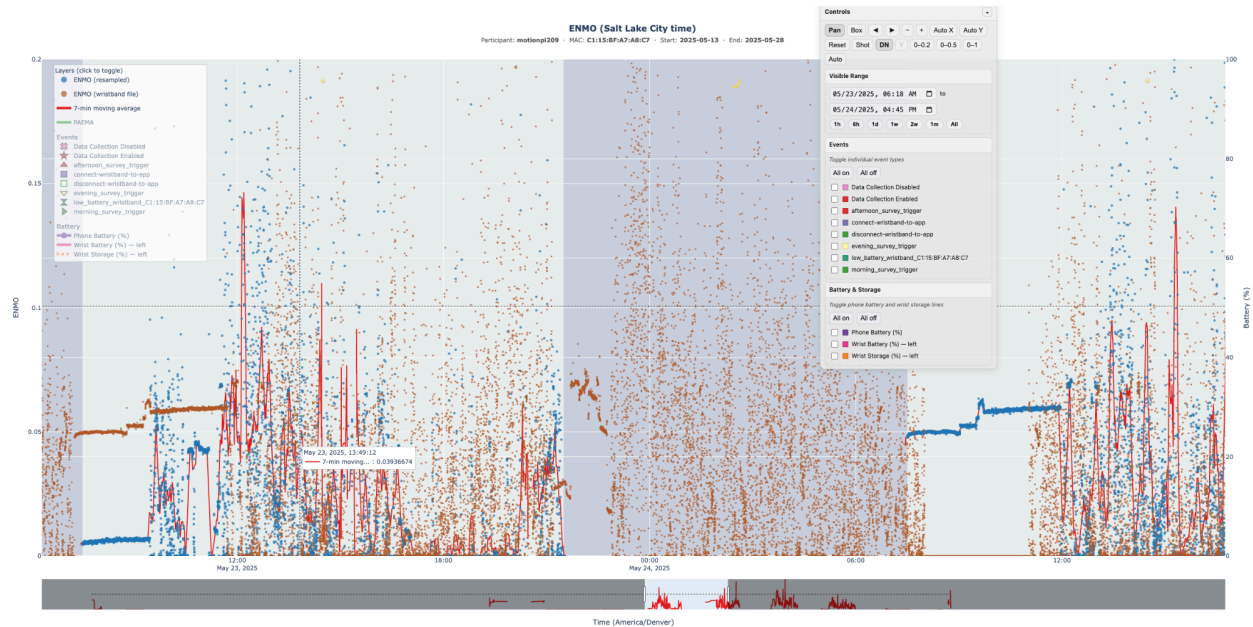
- ENMO points + moving average (orange line)
- Threshold line
- PA-EMA green arrow
- PA-EMA shaded 7-minutes moving green window
- Control panel
- Legend

Why Accepted:

- Supports focus + context + zoom + interactions
- User can toggle layers in legend
- Scales automatically stay readable

7. Final Design (Screenshots Described)

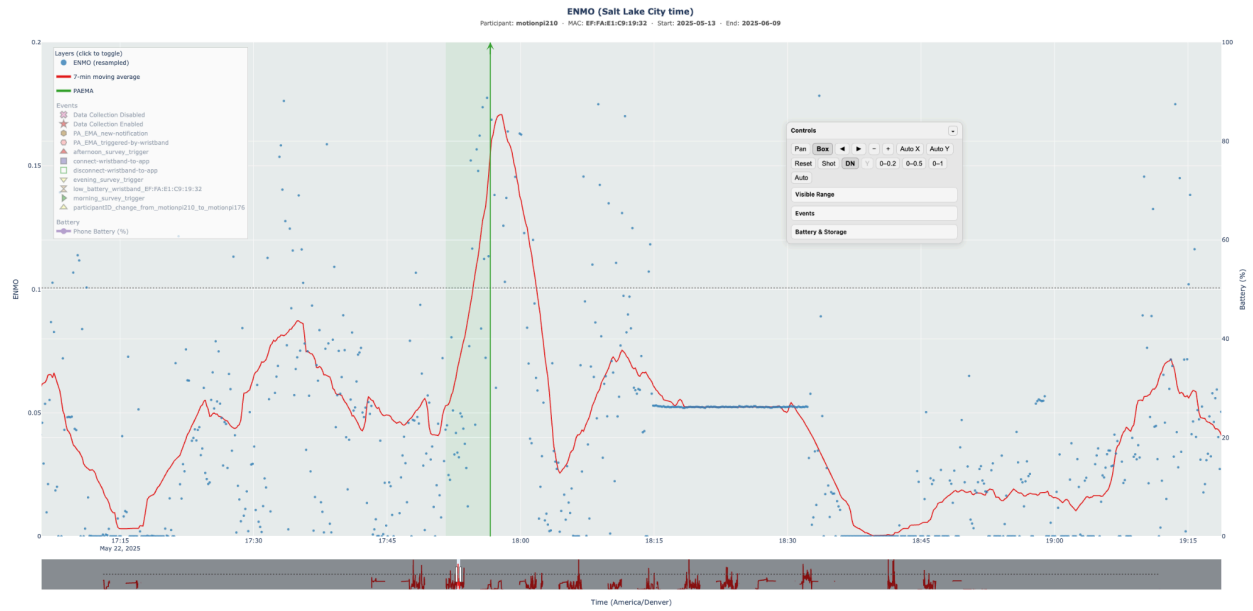
Below are descriptions of the final plot:



A. Full Timeline View

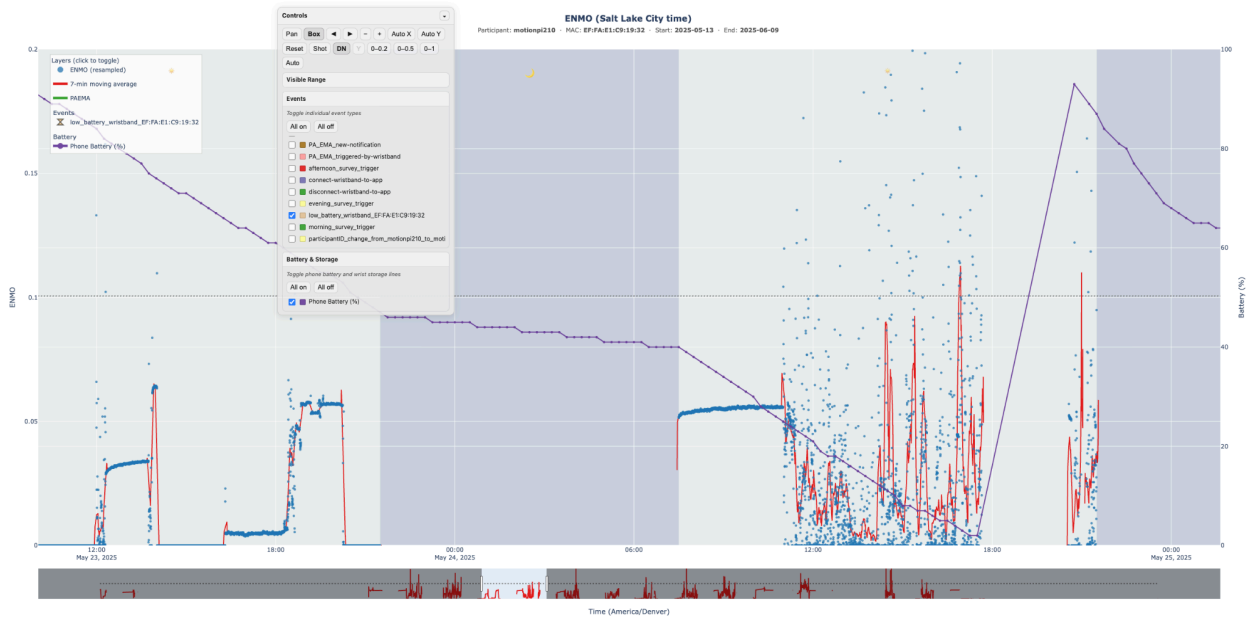
- ENMO from server scatter in blue
- ENMO from wristband storage scatter in red
- Moving average red curve to smooth the signal
- PA-EMA band in green
- Day/night shading
- Reliability metrics on right-axis
- Event markers (triangle, diamond, etc.)

B. Zoomed PA-EMA Window



- Clear visualization of the green 7-minute window
- Trigger arrow
- Raw ENMO vs moving average vs threshold shown clearly

C. Battery Reliability View



- Battery low shown in purple scurve

D. Control Panel

Controls

PanBox◀▶-+Auto XAuto Y

ResetShotDNY0-0.20-0.50-1

Auto

Visible Range

05/24/2025, 04:45 PM📅to

05/27/2025, 12:04 PM📅

1h6h1d1w2w1mAll

Events

Toggle individual event types

All onAll off

☐ Data Collection Disabled

☐ Data Collection Enabled

☐ afternoon_survey_trigger

☐ connect-wristband-to-app

☐ disconnect-wristband-to-app

☐ evening_survey_trigger

☐ low_battery_wristband_C1:15:BF:A7:A8:C7

☐ morning_survey_trigger

Battery & Storage*Toggle phone battery and wrist storage lines*

All onAll off

☐ Phone Battery (%)

☐ Wrist Battery (%) — left

☐ Wrist Storage (%) — left

- Toggle layers
- Y-axis presets

- Range presets
- PNG export

These views illustrate clarity, accessibility, and low clutter.

8. Implementation

The final implementation uses:

Front-End

- Plotly WebGL scatter for >100k points
- HTML + JavaScript inside `src/index.html`
- CSS for layout
- Custom legend + control panel
- Dynamic toggles injected post-render

Performance Features

- WebGL for smooth pan/zoom
- Downsampling for extreme ranges
- Event markers on overlay axis prevent overlap
- Cached moving averages
- Auto-matching left/right wristbands via MAC

Reliability

- No console errors
- Works on Chrome, Safari, and mobile devices
- Handles corrupted timestamps gracefully

9. Evaluation

This section answers whether the tool achieves its goals.

Task Success

Users can now:

- verify PA-EMA windows,
- align event markers with ENMO patterns,
- diagnose reliability issues,
- confirm whether data gaps match battery drops or toggles.

Each task takes < 3 interactions (zoom + toggle layers).

Usability

Beta testing with collaborators showed:

- Clean interpretation
- Shaded windows help non-experts
- Control panel reduces clutter
- Day/night shading improves contextual interpretation

Performance

- Smooth pan/zoom with ~100k points
- No noticeable lag due to WebGL scatter
- Legends and symbols remain readable

10. Limitations & Future Work

Limitations

- Large CSVs (>50MB) may require chunking
- Some wristband timestamp drift is handled heuristically
- Not yet validated with >10 participants
- Hard-coded day/night depends on local timezone

Future Enhancements

- drag-and-drop CSV upload
- multi-participant dropdown

- anomaly detection for long gaps
- improved legends and tooltips
- save/load user view presets
- Machine learning on the plot

11. Conclusion

This project successfully integrates multiple data streams into one interactive visualization, enabling quick analysis of participant behavior, sensor reliability, and EMA trigger validity. The final design is driven by clarity, minimal clutter, and accessibility, ensuring that collaborators from multiple disciplines can interpret the data without requiring programming knowledge.